VAIGAINATHAN R

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SKILLS

Languages: Python, Embedded C, HTML5, CSS3, JavaScript (ES6), PHP, SQL (MySQL)

Frameworks: Flask, Django, Bootstrap

Data Science: NumPy, Pandas, Seaborn, Matplotlib, Power BI, Excel (Advanced)

Tools: VS Code, PyCharm, Git, AWS (S3/EC2), Figma, Adobe Illustrator

Other: Computer Vision, IoT, Embedded Systems, Fusion 360

WORK EXPERIENCE

Software Developer Sep 2025 – Present

Makelabs, Bangalore

- Spearheaded software logic for an AI-powered humanoid robot, improving system response time by 20%.
- Engineered Python-based modules for real-time sensor data processing, handling 50+ signals per second.
- Integrated Embedded Systems with IoT modules to enable autonomous decision-making.
- Developed automation scripts that reduced robot calibration time, saving 10 hours weekly.

INTERNSHIPS

Python Full Stack & Data Analytics Intern Feb 2024 – Aug 2024

PySpiders, Bangalore

- Developed full-stack web applications using Flask and JavaScript, improving user engagement by 15%.
- Optimized MySQL database schemas and complex SQL queries, reducing data retrieval time by 25%.
- Performed exploratory data analysis (EDA) using Pandas and NumPy to identify workflow bottlenecks.
- Designed interactive Power BI dashboards for KPI monitoring and management insights.
- Collaborated in an Agile environment, participating in daily stand-ups and code reviews.

Python Intern Sep 2023 – Nov 2023

YBI Foundation (Remote)

- Built Python automation scripts for bulk file processing, reducing manual effort by 5 hours per week.
- Implemented data cleaning pipelines to prepare datasets for machine learning models.
- Refactored legacy scripts, improving execution efficiency by 20%.
- Developed mini-projects using file handling and exception management.
- Documented workflows to ensure smooth collaboration in a remote team.

PROJECT WORK

Face Attendance Recognition System | Python, AI, OpenCV, SQLite

- Developed an automated attendance system using OpenCV-based face recognition with 98% accuracy.
- Implemented the LBPH algorithm for robust recognition under varying lighting conditions.
- Processed and trained on 1,000+ facial images using a custom preprocessing pipeline.
- Integrated SQLite database to store attendance logs with timestamps.
- Optimized system performance to achieve real-time recognition at 30 FPS.

Vehicle Count System | Python, Flask, Computer Vision

- Built a traffic analysis system using background subtraction and centroid tracking algorithms.
- Developed a Flask backend with RESTful APIs for real-time vehicle count updates.
- Assigned unique IDs to vehicles to prevent duplicate counting across frames.
- Designed a responsive dashboard to visualize traffic density and peak hours.
- Improved detection accuracy by 18% through optimized ROI and noise reduction.

Object Detection System | Python, OpenCV, YOLOv8

- Implemented a real-time object detection system using YOLOv8 supporting 80+ object classes.
- Utilized NumPy-based vectorized operations to increase frame processing speed by 25%.
- Integrated OpenCV DNN module to run pre-trained models on live video feeds.
- Tuned Non-Maximum Suppression (NMS) for accurate bounding box filtering.
- Validated detection reliability in dynamic and high-speed environments.

RFID Attendance System | ESP32, MFRC522, Embedded C, IoT

- Designed a contactless attendance system using ESP32 and MFRC522 RFID module.
- Programmed embedded firmware using SPI communication for real-time card detection.
- Transmitted attendance logs via Wi-Fi to a centralized cloud dashboard.
- Integrated OLED display for instant feedback with sub-100ms latency.
- Implemented deep sleep mode, improving power efficiency by 35%.

EDUCATION

Bachelor of Engineering 2021 – 2025
Jayam College of Engineering and Technology CGPA: 8.1

Higher Secondary Certificate (HSC) 2020 – 2021
Selva Higher Secondary School, Bargur Percentage: 84%

Secondary School Leaving Certificate (SSLC) 2018 – 2019
Vailankanni Matric Higher Secondary School, Bargur Percentage: 81%

AWARDS AND CERTIFICATIONS

- Full Stack Development Certification – Udemy
- Data Science Foundations – Simplilearn
- Academic Excellence Award – Institution of Engineers (India)
- NCC B & C Certificate holder